EX.No.2.1 CREATE A 3D SOLID MODEL FROM ISOMETRIC DRAWING

AIM

To create a 3Dmodel from the given isometric drawing using fusion 360

tool.

REQUIREMENTS

Autodesk fusion 360

Windows 10

16 GB of RAM (integrated graphics recommend 1 GB or more)

2.5 Mbps or faster download; 500 Kbps or faster upload

PROCEDURE

STEP-1: Start the sketch command

Select Sketch > Create Sketch.

STEP-2: Select the Sketch Plane

Select the Front (XZ) Plane

STEP-3: Create a Rectangle

Select Sketch > Rectangle > Center Rectangle.

Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.

Click once to begin placing the rectangle.

Type the Tab button and enter the values 100 mm and 60mm for the width and height respectively.

Click again to place the rectangle.

Click Finish Sketch after completing.

STEP-4: Extrude Rectangle in XZ Plane

Select Extrude.

Select the Rectangle sketch.

Click again to place the value.

Type in the value of 30mm.

Press ENTER to accept the value.

STEP-5: Start the sketch command

Select Sketch > Create Sketch.

STEP-6: Select the Sketch Plane

Select the FRON SURFACE of rectangular block

STEP-7: Create a CIRCLE

Select Sketch >CIRCLE > CENTER CIRCLE.

Now hover over the origin (center) of the sketch. You should see the cursor snap to this location.

Click once to begin placing the CIRCLE.

Type the Tab button and enter the value DIA 100mm .

Click again to place the circle.

Click Finish Sketch after completing.

STEP-8: Extrude CIRCLE

Select Extrude.

Select the Rectangle sketch.

Click again to place the value.

Type in the value of 20mm.

Press ENTER to accept the value.

STEP-9: Start the sketch command

Select Sketch > Create Sketch.

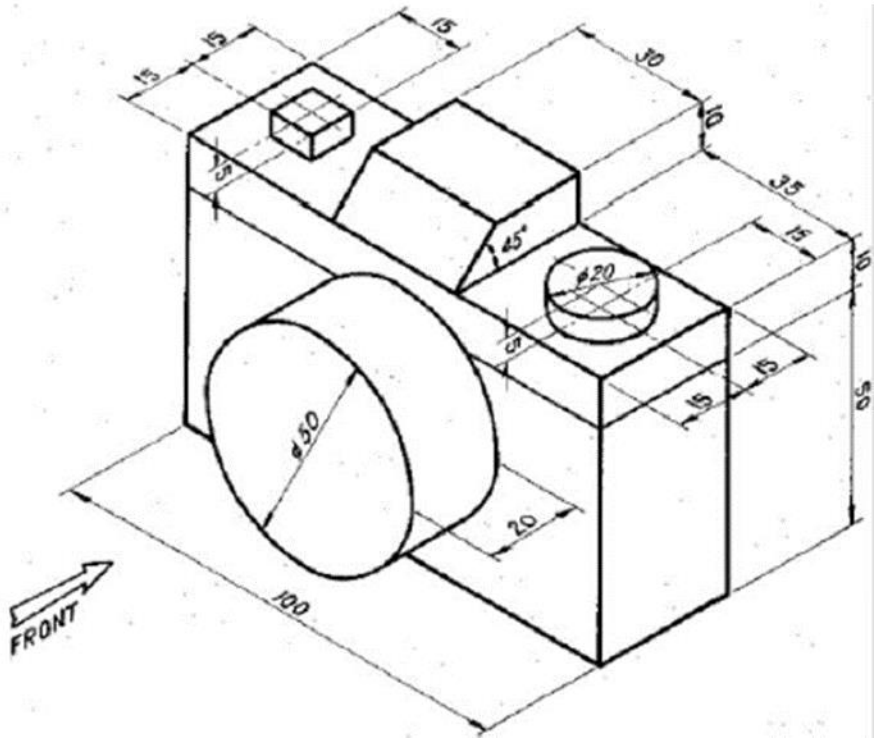
STEP-10: Select the Sketch Plane

Select the TOP SURFACE OF RECTANGLE.

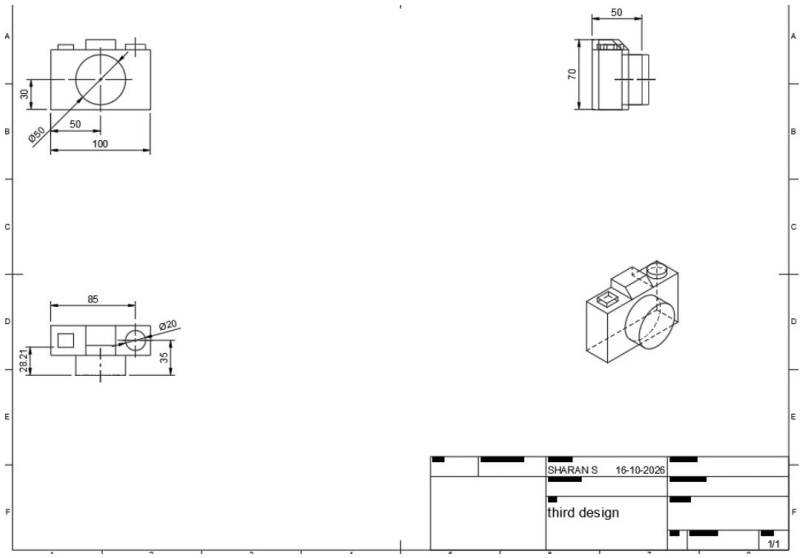
STEP-11: Create a line

1. Select Line command.
2. Draw the projections in the diagram
3. Click Finish Sketch.
4. extrude the sketch.

INPUT


15
15
15
30
10
35
15
10
5
45°
ø20
5
15
15
50
ø50
20
100
FRONT

OUTPUT


50
70
30
50
100
ø50
85
ø20
28.21
35
SHARAN S 16-10-2026
third design
1/1

RESULT:

Thus, the 3D model is created from given isometric drawing.